

WAINWRIGHT, AK, USA

WMO: 700300

Lat: 70.639N

Lon: 159.995W

Elev: 9

StdP: 101.22

Time Zone: -9.00 (NAK)

Period: 99-19

WBAN: 27503

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-38.7	-36.7	-37.3	0.1	-33.4	-36.5	0.1	-33.1	16.2	-16.1	14.9	-16.3	2.5	140	1.063

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.2	17.3	12.7	15.0	11.6	12.9	10.4	13.3	16.4	11.9	14.5	10.6	13.0	5.3	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.7	8.6	15.0	10.1	7.7	13.5	8.8	7.0	12.0	37.5	16.3	33.8	14.7	30.6	13.1	18.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.0	12.2	11.1	-40.1	22.0	3.5	2.9	-42.6	24.2	-44.7	25.9	-46.7	27.5	-49.2	29.7
			-35.5	15.7	1.2	1.6	-36.4	16.9	-37.1	17.9	-37.8	18.8	-38.6	19.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-9.2	-23.6	-22.8	-23.6	-16.0	-4.4	4.0	7.6	6.1	2.2	-5.3	-14.5	-20.5
	DBStd	13.12	7.47	8.65	7.36	6.19	5.32	3.80	3.92	3.63	3.32	4.36	6.43	6.59
	HDD10.0	7027	1043	917	1041	779	445	185	96	131	236	474	736	946
	HDD18.3	10032	1301	1151	1299	1029	703	431	333	378	485	732	986	1204
	CDD10.0	38	0	0	0	0	0	5	21	11	1	0	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	2	0	0	0	0	0	0	2	1	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	5.4	5.1	5.7	5.3	5.5	5.6	5.3	5.2	5.1	5.2	5.7	5.6	5.4
Precipitation	PrecAvg	252	12	12	11	8	9	13	34	53	36	32	16	13
	PrecMax	510	39	35	79	26	34	46	71	115	128	73	71	46
	PrecMin	126	0	2	1	0	1	1	9	18	8	8	4	2
	PrecStd	86	10	9	14	5	8	9	16	25	25	16	12	10
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.9	0.8	1.7	2.2	9.1	19.0	21.9	18.6	13.0	4.1	0.8	-1.3
		MCWB	-1.3	-1.1	-1.3	1.2	6.2	13.1	15.2	14.2	10.0	3.5	0.0	-1.9
	2%	DB	-3.8	-2.3	-3.8	-1.2	5.3	15.3	18.0	15.9	10.9	2.5	-0.6	-3.8
		MCWB	-4.3	-3.2	-4.7	-1.9	3.7	11.3	13.0	12.3	8.8	2.0	-1.3	-4.4
	5%	DB	-8.9	-6.0	-7.7	-3.8	3.1	12.4	16.1	13.3	8.9	1.0	-2.5	-7.7
		MCWB	-9.2	-6.5	-8.1	-4.4	2.0	9.4	12.1	11.2	7.7	0.1	-3.2	-8.0
	10%	DB	-13.6	-9.8	-13.3	-6.8	1.9	9.9	13.8	11.8	7.3	-0.2	-5.0	-11.2
		MCWB	-14.1	-10.3	-13.9	-7.2	1.0	7.6	10.9	10.2	6.4	-0.9	-5.7	-11.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.1	-0.5	-0.2	1.2	6.4	13.5	15.7	14.7	10.4	3.8	0.3	-1.6
		MCDB	-0.9	0.7	1.9	2.0	9.2	17.8	20.6	17.1	12.7	4.1	0.6	-1.4
	2%	WB	-4.0	-2.9	-4.5	-1.9	3.6	11.4	13.8	12.9	9.0	2.1	-1.2	-4.2
		MCDB	-3.5	-2.1	-3.5	-1.3	5.2	14.9	17.6	14.8	10.6	2.5	-0.6	-3.7
	5%	WB	-8.6	-6.0	-7.9	-4.4	2.1	9.5	12.4	11.4	7.8	0.2	-3.2	-8.1
		MCDB	-8.2	-5.4	-7.4	-3.7	3.1	12.2	15.4	13.2	8.9	0.7	-2.4	-7.8
	10%	WB	-13.3	-9.5	-13.4	-7.2	1.0	7.8	11.0	10.1	6.5	-0.9	-5.7	-11.4
		MCDB	-12.8	-9.0	-12.9	-6.9	1.7	9.8	13.7	11.8	7.4	-0.4	-5.2	-11.0
Mean Daily Temperature Range	5% DB	MDBR	6.9	7.1	6.9	7.1	4.7	6.5	7.2	5.6	4.1	4.5	6.3	6.7
		MCDBR	10.9	10.8	9.3	8.1	5.8	11.4	11.3	8.9	6.8	4.1	6.0	8.8
		MCWBR	10.9	10.4	8.7	7.9	4.6	7.8	7.0	6.0	4.8	3.9	5.8	8.7
	5% WB	MCDBR	10.8	10.7	9.1	8.2	5.8	11.1	10.6	8.1	6.5	4.1	6.2	9.0
		MCWBR	10.9	10.4	8.5	7.9	4.5	7.8	7.0	5.6	4.7	4.0	6.0	8.9
Clear-Sky Solar Irradiance	taub		N/A	0.205	0.240	0.304	0.327	0.324	0.337	0.322	0.290	0.204	N/A	N/A
	taud		N/A	2.019	2.217	2.091	2.164	2.406	2.473	2.558	2.547	2.197	N/A	N/A
	Ebn at Noon		0	506	780	812	843	867	840	814	739	523	0	0
	Edh at Noon		0	37	67	109	116	95	87	71	52	29	0	0
All-Sky Solar Radiation	RadAvg		0.00	0.31	1.88	4.14	5.44	5.88	4.77	2.88	1.42	0.38	0.00	0.00
	RadStd		0.00	0.03	0.11	0.27	0.52	0.42	0.53	0.28	0.12	0.05	0.00	0.00

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (4 neighbors)	+1.46	+2.68	+1.38	N/A	N/A	N/A	-483	-485	N/A	N/A

Nomenclature: See separate page